CO4-AT2 – INDUSTRY PROBLEM SOLVING TASK

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1. Smart Surveillance System Selection

Question: A city surveillance system requires latency below 100 ms, accuracy of at least 85%, and moderate computation. The available models are Viola-Jones (78%, low latency), YOLO (92%, moderate latency), and Deep Learning (95%, high latency). Compare and select the model that satisfies all constraints.

Answer: Viola-Jones has low latency but its 78% accuracy is below the required 85%, so it is unacceptable. Deep Learning has 95% accuracy but requires high computation and has high latency, making it unsuitable for the given hardware constraints. YOLO provides 92% accuracy, moderate latency, and moderate computational requirements. Therefore, **YOLO is the best choice** because it satisfies all the stated constraints.

2. Autonomous Vehicle Detection Logic

Question: A vehicle detection system requires accuracy greater than 90% and response time below 50 ms. Method A has 88% accuracy and 30 ms response time; Method B has 93% accuracy and 70 ms; Method C has 91% accuracy and 45 ms. Compare and select the method.

Answer: Method A fails because its accuracy is only 88%. Method B has sufficient accuracy but its 70 ms response time exceeds the 50 ms limit. Method C provides 91% accuracy and a 45 ms response time, satisfying both requirements. Therefore, **Method C should be selected**.

3. Motion Analysis Decision Problem

Question: A traffic system requires accurate motion direction detection and robustness in dense traffic. Optical Flow is accurate but noisy in dense scenes, while Motion Estimation is stable but less detailed. Compare and select the suitable approach for high-density conditions.

Answer: Optical Flow gives detailed and accurate motion information but becomes noisy when the scene is crowded. Motion Estimation provides less detail but is more stable in dense traffic. Since robustness is a key requirement, **Motion Estimation is more suitable** for high-density traffic conditions.

4. Background Modeling Evaluation

Question: A GMM-based system achieves 90% accuracy in static scenes and 65% accuracy in dynamic scenes. The system must maintain at least 80% accuracy across all scenarios. Evaluate whether GMM alone meets the requirement.

Answer: GMM satisfies the requirement in static scenes because its accuracy is 90%. However, its 65% accuracy in dynamic scenes is below the required 80%. Therefore, **GMM alone does not meet the requirement**. An enhanced or hybrid background-modeling approach would be required to improve performance in dynamic environments.

5. Depth Estimation Strategy

Question: Only one camera is available and moderate depth accuracy is acceptable. Stereo Vision provides high accuracy but needs two cameras, while Structure from Motion provides moderate accuracy using a single camera. Compare and select the feasible approach.

Answer: Stereo Vision cannot satisfy the hardware constraint because it requires two cameras. Structure from Motion can operate with a single camera and provides moderate depth accuracy, which is acceptable for the stated requirement. Therefore, **Structure from Motion (SfM) is the feasible choice**.

Conclusion: The selected solutions are YOLO, Method C, Motion Estimation, an enhanced approach instead of GMM alone, and Structure from Motion. Each decision is based on satisfying the constraints specified in the assessment.